Works
in
Progress

Issue 25

Subscribe

Слова , произнесенные Салони Даттани

Любая болезнь — это провал государственной политики

11 августа 2026 года

Translate

23 минуты

Мало кто знает, насколько медицина продвинулась за время их жизни. Еще меньше людей знают, насколько быстрее она могла бы развиваться.

Медицинские инновации изменили нашу жизнь, но мало кто это осознает. Можно сказать людям, что средняя продолжительность жизни в мире удвоилась за столетие, но они не обратят на это внимания — это просто абстрактная цифра, а цифры не передают, насколько другой была жизнь раньше. Поэтому позвольте мне описать это по-другому.

Если бы в 1950 году вы упали в обморок из-за сердечного приступа, вряд ли кто-то из окружающих знал бы, как вам помочь. Искусственное дыхание и непрямой массаж сердца изобрели бы только через десять лет. Если бы вас доставили в больницу, врач, скорее всего, дал бы вам морфин, велел бы лежать и надеяться на лучшее: считалось, что если сердце перестало биться, то уже ничего не поделаешь.

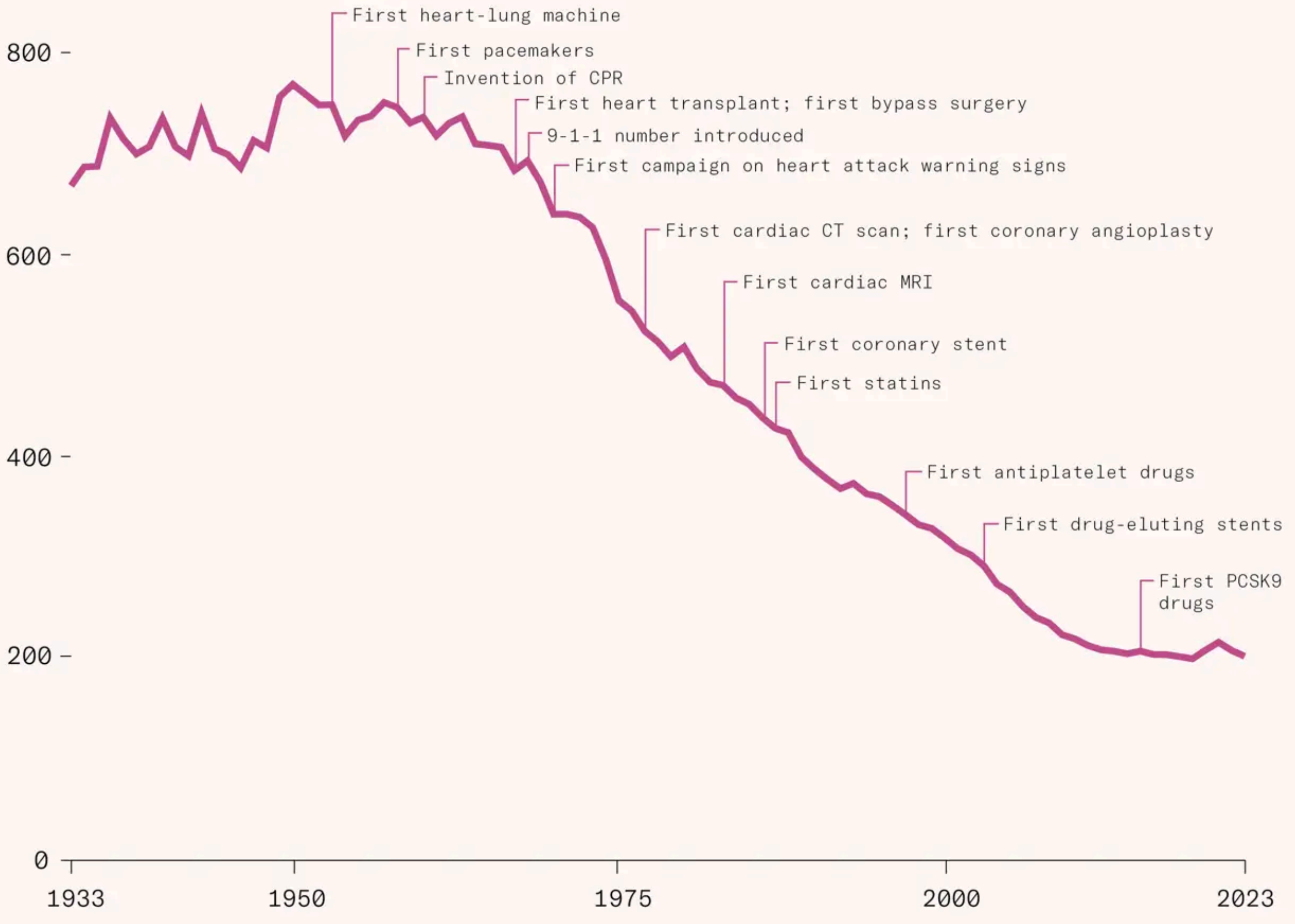
Ситуация изменилась в 1970-х годах. Кардиологи начали понимать, что с каждой минутой все больше сердечной мышцы будет испытывать кислородное голодание, а погибшая ткань не восстановится. Но врачи могли ограничить этот ущерб, быстро восстановив подачу кислорода с помощью тромболитического препарата или баллонного катетера, которые открывали заблокированные кровеносные сосуды и спасали сердечную мышцу.

Другими словами, идее о том, что сердечный приступ можно активно лечить, всего около пятидесяти лет. С тех пор доля пациентов, умерших в течение месяца после перенесения наиболее тяжелого типа сердечного приступа, сократилась на две трети.

Если рассматривать сердечно-сосудистые заболевания в целом, то тенденция еще более драматична. Сегодня у жителей богатых стран вероятность умереть от сердечно-сосудистых заболеваний примерно на четверть выше, чем в 1950-х годах в том же возрасте. Тогда не было ни статинов, ни анализа на холестерин, ни запрета на трансжиры, ни антитабачных кампаний, ни шунтирования, ни имплантируемых кардиостимуляторов.

Cardiovascular death rate in the United States

Age-standardized mortality rate per 100,000



Note: Data begins in 1933, when all US states began reporting cardiovascular mortality rates. Cardiovascular diseases include the ICD codes I00-I09, I11, I13, I20-51, and I60-69.

Source: NCHS (2020); CDC (2025)

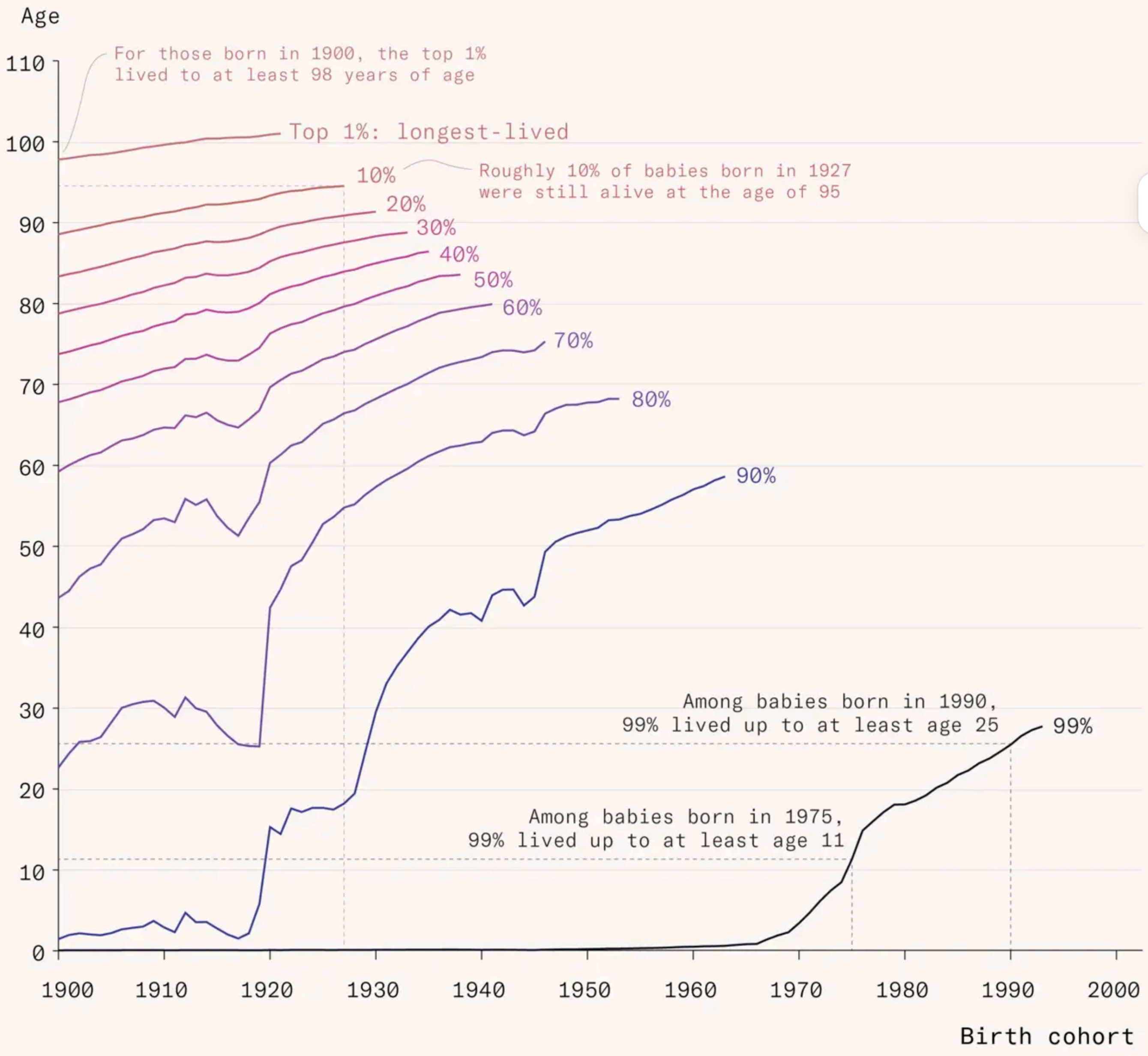
Диабет, некоторые виды рака, инфекционные заболевания и другие состояния стали гораздо более поддающимися лечению, чем раньше. Сто лет назад, если бы у вас был диабет 1-го типа, ваша ожидаемая продолжительность жизни составляла бы несколько месяцев, а «голодная диета» была бы одним из немногих вариантов лечения. Сегодня вам достаточно носить на руке небольшое устройство, которое отслеживает уровень сахара в крови и вводит инсулин для его стабилизации в режиме реального времени. Сто лет назад никто не мог вылечиться от рака, кроме кого-то, в редких случаях — с помощью хирургического ножа. Пенициллин еще не был открыт, и простая раневая инфекция могла привести к сепсису и смерти. Оспа все еще существовала. Она распространялась по всему миру и убивала миллионы людей год за годом.

I think we underestimate how much medicine has progressed because we rarely experience life without it anymore. But it has added years, sometimes decades, to people's lives. Maybe surprisingly, it's also reduced inequality in life expectancy: we've seen greater gains at the bottom of the distribution, among those who would have died as children or young adults. At the same time, life expectancy has risen at every age.

The statistics go like this. In 1950, the shortest-lived 1 percent of babies born in France lived less than three months. But for babies born after that, that 1 percent started surviving longer and longer: up to six months for those born in 1960, three years in 1970, 18 years in 1980, and 25 years in 1990. Alternatively, you can think about the cohort you went to school with. If you were born in France in 1900, only two thirds of your cohort would still be alive with you when you reached the age of sixty. But if you were born in 1960, 90 percent of them would.[1] So here's another way to think about the rise in life expectancy. It means that far fewer parents lose their children, fewer children lose their siblings, and we grow old with more of our friends and family still alive.

Share of each birth cohort that survived until a given age, France

Data is less complete for longer-lived percentiles and more recent cohorts, as many of those people are still alive.



Source: Human Mortality Database (2025); Alvarez & Vaupel (2023); adapted to cohort estimates by Saloni Dattani

A technological revolution

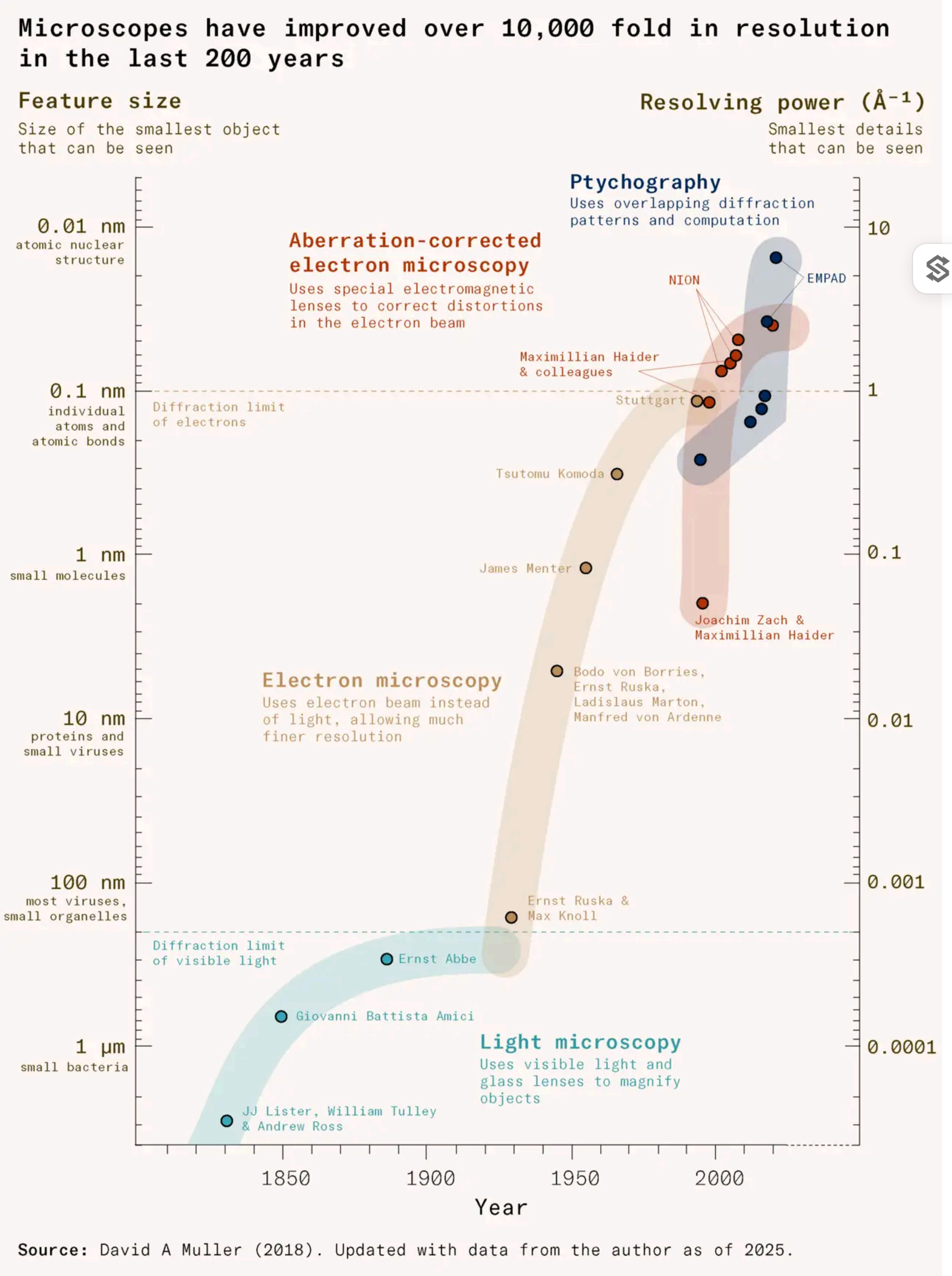
I used to have the impression that medical breakthroughs were uncommon; sometimes happening by pure chance, sometimes by sheer determination. I now see it as a continuous stream of medical innovation every year.

In the last few years, we've had new antiviral drugs against HIV that can prevent infections with nearly 100 percent efficacy, with just a single dose given every six months. There are new drugs that reduce cholesterol levels by 60 percent for people already on statins. There are new treatments that slow the progression of cancers, including certain lung cancers, brain cancer, pancreatic cancer and multiple myeloma, by more than half. And over the past five years, we've had the first vaccines against four more diseases: Covid-19 of course, but also the first malaria vaccine, the first chikungunya vaccine, and the first RSV vaccine.

For most of our history, finding effective new treatments really was a rare occurrence. After Edward Jenner discovered the first vaccine, it took nearly ninety years for scientists to work out how to develop the second. People relied on plants and natural remedies for centuries, with little idea of which ingredients worked or why. Some of these remedies were effective, but contamination was common, and without knowing their active ingredients, they couldn't be produced reliably at scale.

After chemistry matured into a scientific discipline in the nineteenth century, it became possible to isolate and purify compounds, synthesize new ones, and screen hundreds or thousands at once to find those that had medical effects. Chance discoveries could now be made systematically, and chemists discovered medicinal compounds in soil (yielding antibiotics like streptomycin, vancomycin, and tetracycline), fungi (statins and immunosuppressants like cyclosporin), and coal tar and dyes (sulfa antibiotics and early anesthetics like benzocaine). It was a valuable strategy when little was understood about the biological causes of diseases or the mechanisms of drugs.

By the late twentieth century, new tools, theory, and knowledge made it possible to develop treatments based on better biological understanding in an approach that became known as 'rational drug design'. Scientists learned to decipher the structure of the proteins that drove disease and design molecules to fit into them and block them. They began to grow and study living cells in the lab, engineer bacteria and yeast to bulk-produce proteins (including insulin, clotting factors, and antibodies), and automate the screening of hundreds of thousands of compounds in days. It all became faster and possible at a far greater scale. Back when the Human Genome Project was completed in 2003, it cost $50 million and took six months to sequence just one person's genome. Now it costs a few hundred dollars and takes under four hours. The tools have also gotten sharper. Over the past two hundred years, the resolution of the best microscopes has increased over ten-thousandfold. Until the 1930s, no one had ever seen a virus. Now we can see them down to their atoms.



Advances in AI are taking us further. Tools like AlphaFold can help predict the shape of proteins in minutes rather than years. Others are being used to design entirely new proteins that don't exist in nature, including enzymes that catalyze reactions no natural enzyme can and proteins that target cancer cells and destroy disease-causing proteins within them. Some tools can identify hidden microbial genes that encode antibiotics but aren't expressed under standard laboratory conditions. Others have uncovered ancient versions of today's CRISPR enzymes, much smaller than the ones we use now but still able to protect bacteria from viruses. They have turned up entirely new classes of gene-editing tools that can precisely cut and paste long stretches of DNA, allowing them to fix multiple genes at once.

Much of the early research behind these breakthroughs takes place in academic labs and institutes funded by governments and foundations, before they are developed into drugs, manufactured, tested in large clinical trials, and brought to market by pharmaceutical companies. The first antiviral drug against HIV, for example, came from cancer researchers at the National Cancer Institute, who systematically screened hundreds of drugs to test whether any of them inhibited the virus before eventually identifying AZT. Burroughs Wellcome (now part of GSK) then ran the clinical trials to get it licensed. Similarly, the GLP-1 hormone was identified by academic researchers in the 1980s, but it was scientists at Novo Nordisk who developed drugs that mimicked it and chemically modified them to last long enough in the body to be used medically for diabetes and weight loss.

Competitive pressure generally causes firms to underinvest in open, early-stage research: if they publish it, other firms can use their findings without contributing; if they don't, others can't test, challenge or learn from their work. So private firms tend to focus on later-stage research, predominantly for diseases with a large commercial market, as they are better equipped to do the capital-intensive work of running large clinical trials, navigating regulators, and scaling manufacturing than academic institutions and foundations. Together, this process of exploration, discovery, development, and scaling has transformed once-fatal diseases into manageable, and sometimes even curable, conditions.

Medical innovation is one of the most valuable things that happens in a modern economy. Economists estimate that gains in life expectancy in the United States from 1970 to 2000 were worth about $95 trillion to Americans, with roughly half of that coming from reductions in heart disease mortality.

Bottlenecks and broken incentives

There are almost certainly effective treatments in the pipeline today that won't reach patients for years, sometimes decades, and it's not because of the science.

There are few better examples than the malaria vaccine. Malaria is a very complicated disease: it's caused by a parasite that changes shape multiple times during its life cycle, making it very hard to know what to target with a vaccine. So the first malaria vaccine, introduced just a few years ago, was an amazing scientific breakthrough. But it was actually developed in the 1990s. The researchers who developed it struggled to find funding to test it at every stage of the process. There was little commercial incentive. It's not profitable to develop vaccines for diseases that affect people in poorer countries, without being able to recoup the R&D investment – even if millions of children might benefit and there's a huge economic return. It took foreign aid and philanthropy to fund it, and it took literally decades to reach children who needed it, even as half a million of them were dying from malaria every year.

Those children aren't coming back. But we can prevent this from happening again. Economists have come up with an idea to change the incentives with an advance market commitment, where donors promise ahead of time to buy a vaccine at a certain price per dose. The payout occurs only if it's developed, and proven safe and effective. The commitment gives companies the confidence to invest, and can bring vaccines into existence that would otherwise never get made. Most importantly, it makes sure they're manufactured at scale and sold at an affordable price, so they reach those who need them.

That idea was used over a decade ago to develop new vaccines for pneumococcal disease, a deadly bacterial infection of the lungs and brain. Pneumococcal vaccines already existed, but they didn't include the strains common in Africa and South Asia. In 2009, several countries and philanthropists came together to fund an advance market commitment for new vaccines and it succeeded: several companies developed them and they reached children much faster than usual. It's estimated that those vaccines have saved the lives of over 700,000 young children since then.

For all that, malaria is one of the better-funded global health diseases. Roughly $4 billion is spent on R&D across all tropical diseases combined each year, or around six cents for every thousand dollars of income in wealthy countries. Two thirds of it goes toward tuberculosis, malaria, and HIV. For the rest, the research pipeline is incredibly narrow, if it exists at all. One example is trachoma, a painful bacterial infection of the eyes that's spread by flies. Repeated infections scar the eyelids, which gradually turn inward until the eyelashes scratch the eyes. Eventually, the scarring clouds the cornea, causing the person to go blind. Although we've made a lot of progress against new cases of the disease with antibiotics, around 400,000 people worldwide are blind because of the disease, and the pipeline of new diagnostics, treatments and vaccines in development for it is nearly empty, with R&D spending in 2023 falling to zero.

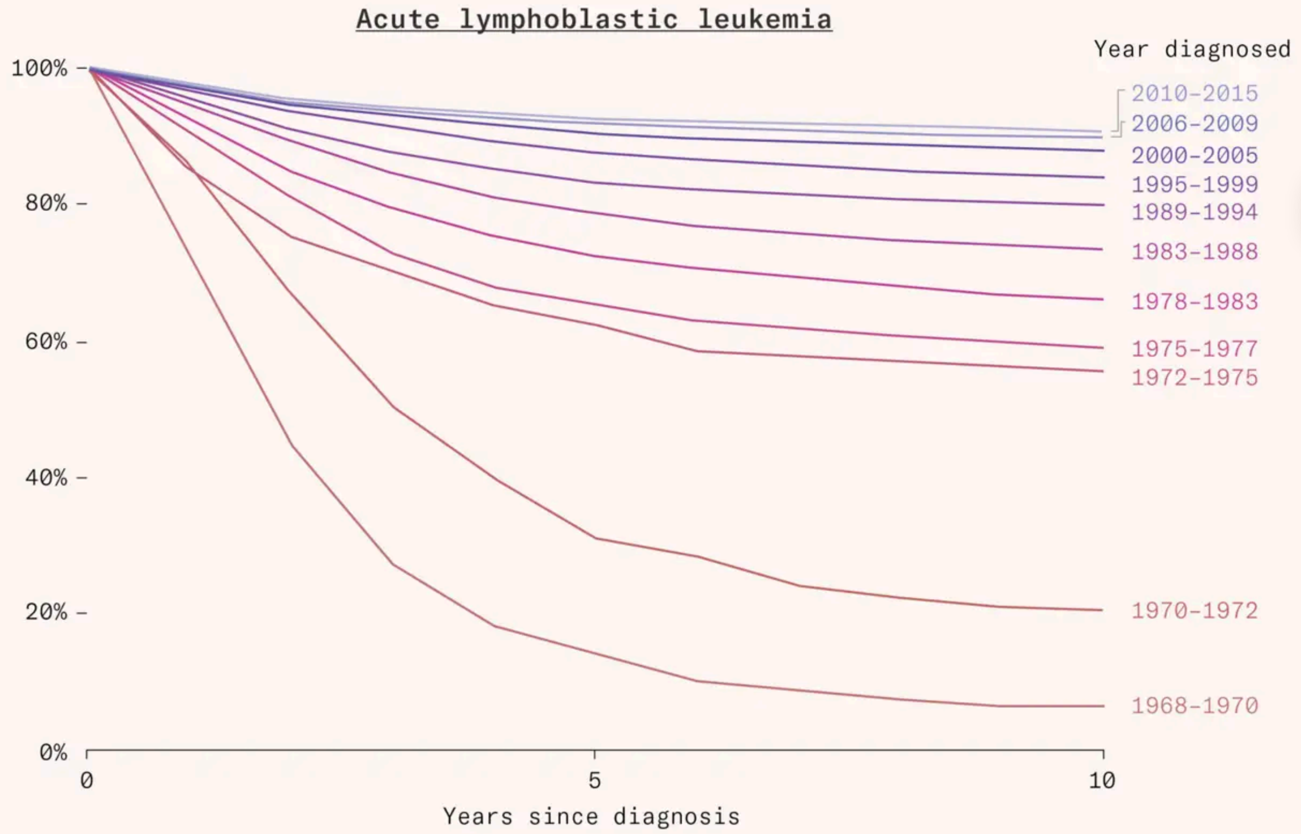
Many other tropical diseases, like leptospirosis, scabies, Buruli ulcer, mycetoma, and yaws, receive almost no R&D funding, and have research pipelines so thin they can support only a few dozen researchers worldwide. For some, like mycetoma, even the basic biology is mysterious, with little understanding of which molecular pathways drive the disease or how to target it with drugs.

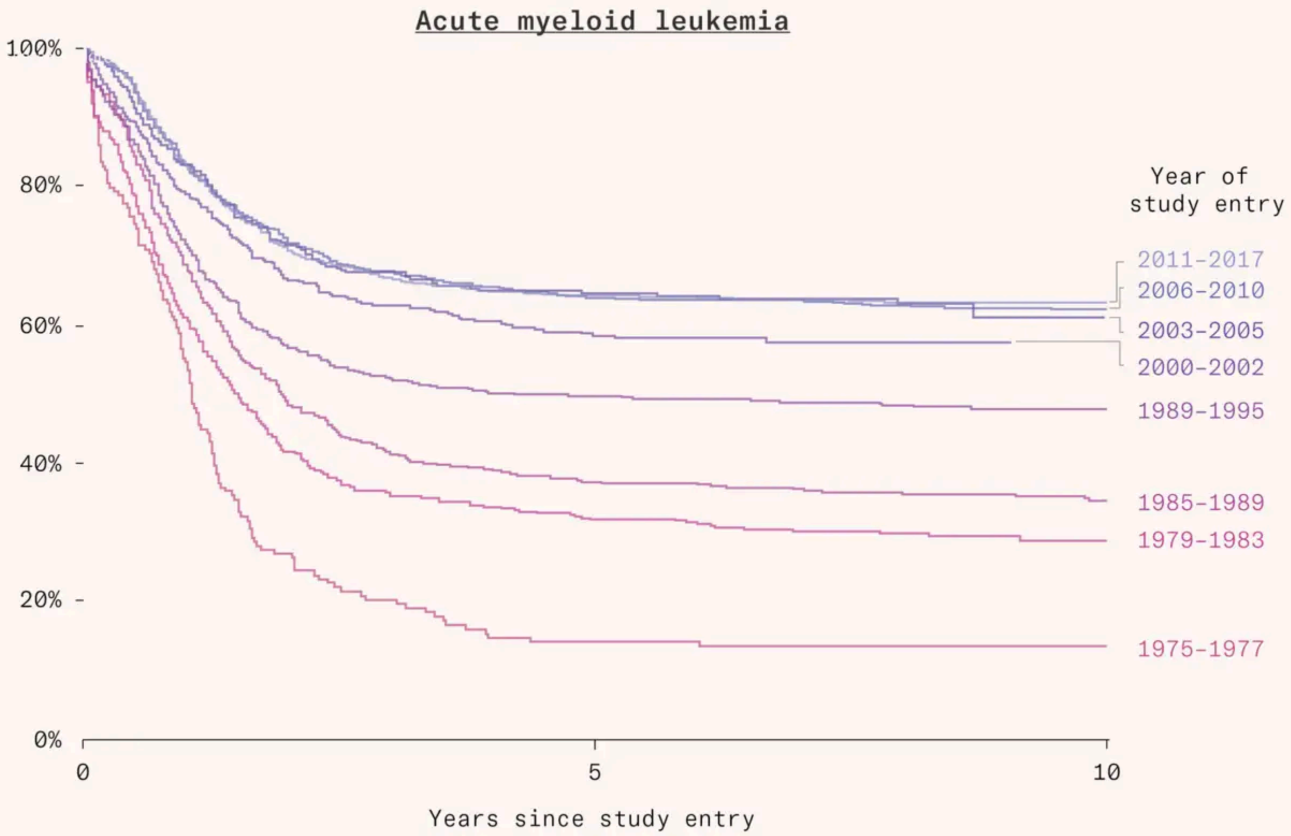
The problem extends far beyond diseases of global poverty. Rare diseases, conditions affecting fewer than 1 in 2,000 people, are another example. Thousands of rare diseases have been described, and they collectively affect up to 6 percent of the population. But each disease is rare enough that the financial market for new drugs is limited and patients are often abandoned entirely; 95 percent of these diseases don't have any approved treatment at all. Running trials for these conditions takes thinking differently: using smaller but more efficient trials, more coordination between researchers, and different approval pathways. For instance, many new gene therapies edit or silence specific harmful genes. Once researchers have found a safe way to get these therapies inside the cells where they're needed, regulatory systems could treat the delivery method as a single platform, allowing the particular genetic sequence to be switched out to target different mutations of the same disease, or even different genetic diseases, without needing new approvals each time.

Or consider childhood leukemia. It used to be very difficult to test treatments for the disease because it is rare, and individual hospitals struggled to find enough patients to run clinical trials. So, from the 1960s onward, researchers built networks to recruit patients from across the US, and later Canada and Europe into larger clinical trials. That collaboration made it possible to learn what worked faster, and it's why leukemia is no longer the disease it used to be. When I used to hear the phrase childhood leukemia, I'd picture a child who suddenly fell ill and whose parents were told they only had a few years left to live. That has changed. Before the 1970s, only 15 percent of children with leukemia would be alive five years after diagnosis. Now that figure is 85 percent. Most children in richer countries today survive and are effectively cured of the cancer.

Share surviving from childhood leukemia

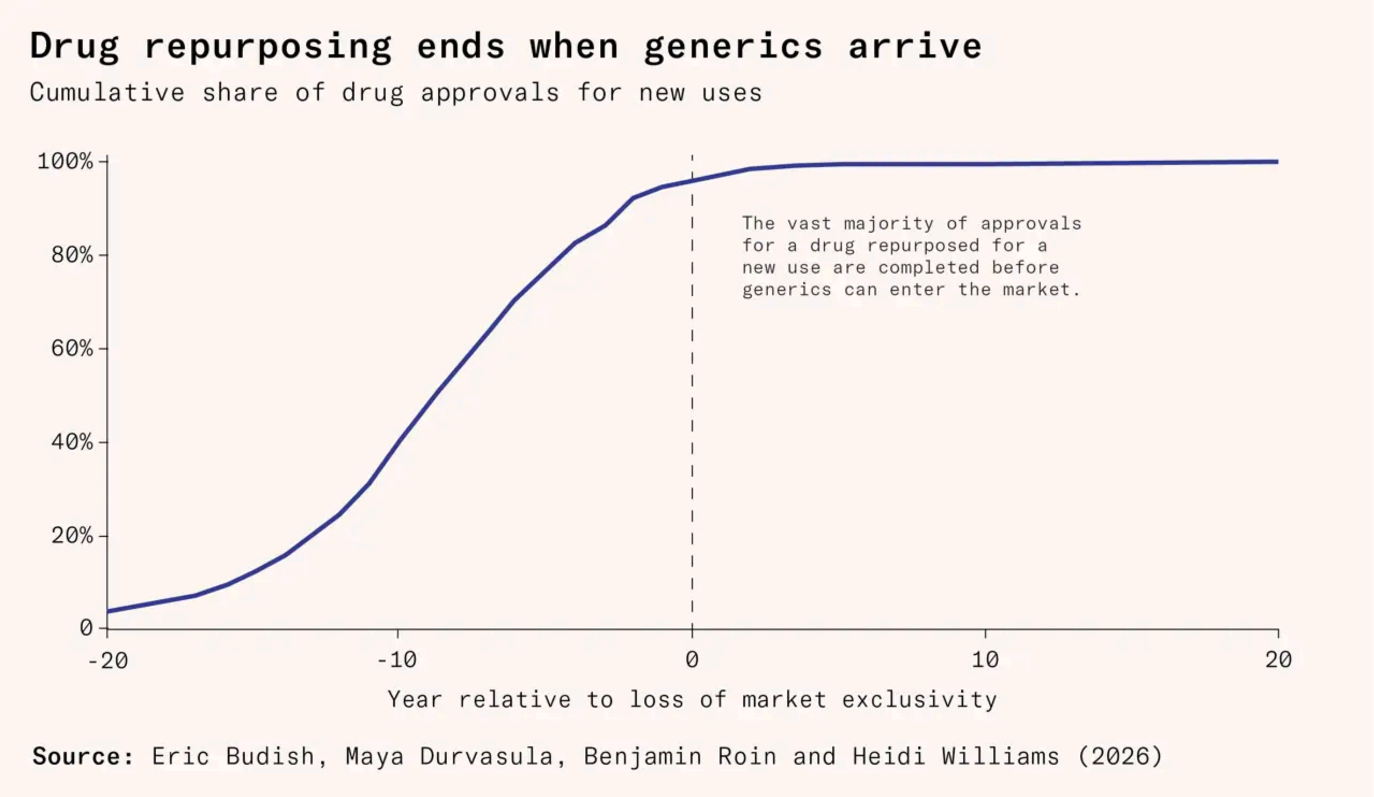
Survival rates for the two most common forms of childhood leukemia: acute lymphoblastic leukemia (80% of cases) and acute myeloid leukemia (20% of cases) in trials across the United States, Canada and other countries.





Source: Children's Oncology Group trials (2023)

These problems fall especially hard on tropical and rare diseases, but poor economic incentives mean we're missing out on medical innovation across the board. Some of the biggest breakthroughs in medicine have been drugs developed for one condition that turned out to be effective for others, like GLP-1 drugs, which were initially approved for diabetes before being used for weight loss. Or SGLT2 inhibitors, which were also originally developed for diabetes, and have since turned out to be effective for treating heart failure and chronic kidney disease as well. This kind of 'drug repurposing', where companies test their own drugs for new uses, can be very profitable, but the incentive is time-limited: once a drug goes off patent and cheaper generics enter the market, the reward for running large expensive trials disappears and firms stop seeking approval for new uses. We don't find out whether the drugs would have worked, and patients who might have benefited can't access them.



Often, we may have many plausible candidates but no cheap way to test them. One way to fix this is through platform trials, which test multiple treatments for a single disease using a shared control group, which lowers the cost of testing each one and makes it possible to compare a range of treatments directly with one another. The RECOVERY trial in the United Kingdom, for example, tested over a dozen drugs as potential Covid treatments in a single trial over two years, at a fraction of the typical cost. Within a few months of the beginning of the pandemic, the trial identified that dexamethasone, a generic drug, was effective against Covid, cutting deaths by around a third among the sickest patients on ventilators. Its use is estimated to have saved hundreds of thousands of lives since then. The trial also ran much faster than usual, because participants were recruited through the UK's National Health Service with simple, streamlined ethical approvals and consent forms.

The deeper problem is that technology isn't always the barrier: sometimes it's about the funding, the incentives, and the institutions. While genome sequencing costs have plummeted and the resolution of microscopes has surged, productivity in drug development has actually slowed down: the cost of getting a drug to market has been doubling in real terms every nine years for decades. This phenomenon was first described by Jack Scannell, who coined it 'Eroom's law', the reverse of Moore's law, which says that transistor chips roughly double in density every two years, allowing smaller and cheaper computers. The slowdown may be part of a broader trend: across many fields, including semiconductors and agriculture, the number of researchers has risen while the output per researcher has fallen and productivity has slowed down.

One explanation is that the lowest-hanging fruit is gone: many medicinal compounds in nature have already been screened, for example, and newer ones take more resources to discover. But given how much our tools of discovery have improved, I'm not convinced. More persuasive, I think, is the 'better than The Beatles' problem: new drugs not only have to be safe and effective, they have to be superior to the current standard of care, meaning they must clear a bar that rises with each generation of successful treatments. This isn't just because older drugs already have some efficacy, but also because many have gone off patent and become generics, meaning they can outcompete new drugs on price as well. And as existing treatments improve and diseases become milder, it becomes statistically more challenging to confirm the benefits of new drugs, requiring larger or longer clinical trials.

It's also become much slower and more complicated to run clinical trials. Before a drug can be approved, it must pass through successive stages of trials, with more participants and more data collected at each stage, before the data is submitted to regulators and approved for medical use. That overall timeline has grown from an average of six years in the 1970s to eight to nine in the 2010s.

Currently, each trial is built from scratch, with its own infrastructure, contracts and administrative overhead. Before a single patient is enrolled, each hospital or research center running the trial must review the protocol, negotiate contracts, obtain ethics approval and train staff; this process takes several months for each trial. Recruiting enough participants is then another challenge: trials seek out patients who meet narrow eligibility criteria, are willing to be randomized to treatment or placebo (meaning they may not receive the treatment at all), and will return to a clinic for follow-up visits sometimes dozens of times over several years. Around half of trials fail to recruit as many participants as planned; only one in five finish on time, with median delays of over a year. On average, it costs $54 million in the US, or around $37,000 per patient, to run a single late-stage trial for a single infectious disease drug.

The bureaucracy that surrounds trials today is largely a legacy of global regulatory guidelines introduced in the mid-1990s, which the trials industry interpreted defensively to minimize the risk of regulatory rejection. Even when regulators have since rolled back specific requirements, companies have been slow to change. Around a third of trial costs, for example, come from the practice of 'source data verification', where a consultant manually cross-checks every data point collected at trial sites against paper records, despite the FDA recommending against it because it catches few errors while adding huge costs. Because less than 10 percent of drugs that enter clinical trials are approved and the costs of development are large, the industry has remained financially risk averse, doing far more administrative work than is required, and far less efficiently than the process could allow.

That doesn't mean we should sidestep the process of rigorous testing. Clinical trials are the best way to test whether drugs are actually effective, how effective they are, and what side effects they carry. Without them, we can't reliably identify which drugs actually work, and we can easily be misled by random fluctuations, natural improvements over time, and selection biases that make drugs appear effective even if they have no benefit or actively harm patients. But the whole process could be much more efficient. It's possible to speed up medical innovation without cutting corners or compromising on rigor, or better, through win-win solutions that improve both.

A cure for every disease

Far too many diseases are still untreatable. It’s painful to imagine gradually forgetting your children’s names, forgetting how to hold a conversation, how to recognize your own home, and eventually even how to swallow food, while your family watches someone they love disappear. But that’s the reality for over thirty million people with Alzheimer’s globally, and we have almost nothing to offer them. Part of the reason may be because the brain is so complex, or because it lined by the blood-brain barrier, a tightly sealed layer of cells which keeps out most pathogens and toxins, as well as most drugs. That makes it challenging to design a drug that can cross the barrier, reach the right cells, and do what it’s meant to do without any unexpected side effects.

Cancer is not a single condition but hundreds of distinct diseases, driven by different genetic mutations, that can evolve rapidly and become resistant to each drug. Treating them often requires combination treatments, which attack cancers from multiple angles to reduce the chances of resistance developing, but many cancer drugs carry side effects that are highly unpleasant or even debilitating. Often, the only available options are chemotherapy or radiation therapy, which attack healthy tissue alongside tumors, causing lasting damage to the body.

I’m hopeful that progress is possible. Many conditions that once seemed untreatable have turned out not to be. Hepatitis C is now curable. Some diseases are now entirely preventable with vaccines, including cervical cancer and many liver cancers. Multiple liver diseases can be halted by new drugs that silence key proteins that drive their progression. CAR-T cell therapies, which reengineer a patient’s own immune cells to kill their cancer cells, mean that several blood cancers can now be put into lasting remission, and sometimes cured. New treatments have given cystic fibrosis patients lives that look much like anyone else’s, when they would have otherwise died in their twenties or thirties.

The most dramatic example is pancreatic cancer, one of the deadliest cancers, which has had a five-year survival rate of just 13 percent. It’s long been thought to be untreatable, because it is driven by mutations in the KRAS protein, which is notoriously hard for drugs to target because the surface of the protein is smooth, without an obvious pocket for drugs to bind to and block. In the last decade, however, breakthroughs in understanding the protein’s structure and interactions have helped develop new drugs that recruit a second protein to grab onto KRAS and block it. A new large phase three trial found that one such drug, daraxonrasib, roughly doubled survival compared to standard chemotherapy. Its side effects are rough, but knowing how to target the protein at all means scientists can build upon it. That work will be slow and difficult, but decades from now, we'll look back and see the trajectory has changed, as it did for leukemia.

The list of bottlenecks is long, and we need to move faster if we want to solve them before it's too late for too many people. It takes years to find and recruit patients for clinical trials and clinical trial infrastructure is highly fragmented. Commercial incentives mean that firms have little desire to develop drugs for many diseases whose patients are too few or too poor for it to be profitable, and there's little incentive to repurpose drugs for new uses once patents run out. People have already come up with some ideas to solve them, including platform trials, advance market commitments, prizes to incentivize drugs for neglected diseases, risk-based monitoring of trials to focus on the highest-risk elements rather than on cross-checking every data point manually, large simple trials that operate through existing electronic health records, and continued randomization after a drug is approved to learn which dosing strategies, treatment durations, and drug combinations work best. But we also need new ideas, and people to test them, scale them up, and see them through.

We can make progress even in the hardest of cases. Ebola vaccines are incredibly challenging to develop through the regular route for many of the same reasons. There was little commercial incentive: Ebola largely affects the poorest, and its outbreaks are unpredictable, which makes it hard to set up trials in time. But what's amazing is that we have a vaccine for the Ebola Zaire virus. Just over a decade ago, Gavi created an advance market commitment to fund it, and scientists came up with a different way to run clinical trials: ring vaccination. They waited for individual cases of Ebola to appear, and then quickly vaccinated the contacts of affected people, testing how well the vaccine prevented new infections. The World Health Organization coordinated the trials, the US government helped fund the vaccine's development through BARDA and the Department of Defense, and once the vaccine was licensed in 2019, Gavi, WHO, UNICEF, and the Red Cross set up a global stockpile so doses would be ready for another outbreak. That stream of innovation, I've realized, is the result of countless people who thought differently, whether they were scientists, economists, managers or operators. Some of them changed their lives to make it happen. What they recognized was something quite simple: that diseases are not a fact of life, but problems that can be solved.

I've been writing about global health and medical innovation for years, and when I tell friends and family how much progress there's been, the reaction is almost always the same: they had no idea most of it was happening. But progress isn't inevitable, and breakthroughs don't just come about when we have the tools to make them; they depend on how innovation happens in the real world. Right now, those processes are bottlenecked by funding, institutions and incentives. But we can change them. I think it's more important than ever that we do so, because of cuts to spending on science, global health and foreign aid; we need to find ways to make our resources go further.

The central drama of the modern world is innovation against suffering, and we have made incredible progress so far. But we are doing far worse than we could. Millions still suffer from untreatable diseases, because of bottlenecks that we could fix. Fixing them is one of the most important things we can do. You can help make it happen.

1

These calculations are conditional: they ask what share of the children alive at age 5 ('the cohort you went to school with') went on to reach age 60.

Saloni Dattani is an editor at Works in Progress, writes the newsletter Scientific Discovery, and is an advisor to Coefficient Giving on clinical trial reform.

This piece was adapted from a TED talk by the author.

Get the print magazine

Subscribe for $100 to receive six beautiful issues per year.

Subscribe

Similar pieces

Advancing antivenom

Words by **Mathias Kirk Bonde**

Snakebites kill between 80,000 and 140,000 people every year. Better antivenom should be a high priority – thankfully new technology can help.

Read more →

Biotech

The kill switch for harmful genes

Words by **Jacob Witten**

What if instead of editing a gene, or blocking the protein it produces, you could simply switch it off?

Read more →

Biotech

Epidurals are a miracle technology

Words by **Phoebe Arslanagic-Little**

Childbirth may be the most painful thing a person can experience, but we have the technology to numb the pain.

Read more →

Biotech

Contact

Contact
worksinprogress@stripe.com

Questions about your subscription?
wip-print@stripe.com